a. Show that 3-SAT belongs to the class NP;

b. Reduce the CNF-SAT problem to 3-SAT;

c. Deduce that 3-SAT is NP-Complete.

Solution：

a. Show that 3-SAT belongs to the class NP;

A certificate for a "yes" instance of 3-SAT is a **satisfying assignment** of the variables.

Given a 3-CNF formula ϕ and a candidate assignment, the verifier checks whether the assignment satisfies ϕ. This involves:

- Evaluating each clause in ϕ under the given assignment.
- Ensuring that **at least one literal** in each clause is true.
- If all clauses are satisfied, the verifier accepts; otherwise, it rejects.

Let ϕ have m clauses.

Each clause has exactly 3 literals, so checking a clause takes $O(1)$ time.

Thus, checking all m clauses takes $O(m)$ time, which is **polynomial in the input size**.

Since the verifier runs in polynomial time, **3-SAT is in NP**.

b.

Given a CNF formula ϕ, we process each clause C_i as follows:

1. **If C_i has exactly 3 literals**, leave it unchanged.
   Example: $(x_1 \vee \bar{x}_2 \vee x_3)$ remains the same.
2. **If C_i has 1 literal**, say l_i, replace it with:
   $(l_i \vee y_{i1} \vee y_{i2}) \wedge (l_i \vee \bar{y}_{i1} \vee y_{i2}) \wedge (l_i \vee y_{i1} \vee \bar{y}_{i2}) \wedge (l_i \vee \bar{y}_{i1} \vee \bar{y}_{i2})$
3. **If C_i has *2* literals**, say $l_{i1} \vee l_{i2}$, replace it with:
   $(l_{i1} \vee l_{i2} \vee y_i) \wedge (l_{i1} \vee l_{i2} \vee \bar{y}_i)$
4. **If C_i has *k* literals (*k*>3)**, say $l_{i1} \vee l_{i2} \vee \cdots \vee l_{ik}$, replace it with:
   $(l_{i1} \vee l_{i2} \vee y_1) \wedge (\bar{y}_1 \vee l_{i3} \vee y_2) \wedge \cdots \wedge (\bar{y}_{j-2} \vee l_{ij} \vee y_{j-1}) \wedge \cdots \wedge (\bar{y}_{k-4} \vee l_{ik-2} \vee y_{k-3})$
   $\wedge (\bar{y}_{k-3} \vee l_{ik-1} \vee l_k)$

2. **If C_i has 1 literal**, say l_i, replace it by Z_i :

$(l_i \vee y_{i1} \vee y_{i2}) \wedge (l_i \vee \bar{y}_{i1} \vee y_{i2}) \wedge (l_i \vee y_{i1} \vee \bar{y}_{i2}) \wedge (l_i \vee \bar{y}_{i1} \vee \bar{y}_{i2})$

I II III IV

Truth Table :

l_i	$y_{i,1}$	$y_{i,2}$	I	II	III	IV	Z_i
T	T	T	T	T	T	T	T
T	T	F	T	T	T	T	T
T	F	T	T	T	T	T	T
T	F	F	T	T	T	T	T
F	T	T	T	T	T	F	F
F	T	F	T	F	T	T	F
F	F	T	T	T	F	T	F
F	F	F	F	T	T	T	F

When C_i (i.e. l_i) is true, the new Z_i is true
When C_i (i.e. l_i) is false, the new Z_i is false
Hence C_i can be reduced to Z_i where each clause Z_i contains exactly 3 literals and
$C_i \Leftrightarrow Z_i$

3. **If C_i has 2 literal**, say $l_{i1} \vee l_{i2}$, replace it by Z_i :

$(l_{i1} \vee l_{i2} \vee y_i) \wedge (l_{i1} \vee l_{i2} \vee \bar{y}_i)$

I II

Truth Table :

$l_{i,1}$	$l_{i,2}$	y_i	C_i	I	II	Z_i
T	T	T	T	T	T	T
T	T	F	T	T	T	T
T	F	T	T	T	T	T
T	F	F	T	T	T	T
F	T	T	T	T	T	T
F	T	F	T	T	T	T
F	F	T	F	T	F	F
F	F	F	F	F	T	F

When C_i is true, the new Z_i is true
When C_i is false, the new Z_i is false
Hence C_i can be reduced to Z_i where each clause Z_i contains exactly 3 literals and
$C_i \Leftrightarrow Z_i$

4. If C_i **has *k* literals (*k*>3)**, say $l_{i1} \vee l_{i2} \vee \cdots \vee l_{ik}$, replace it by Z_i :

$(l_{i1} \vee l_{i2} \vee y_1) \wedge (\bar{y}_1 \vee l_{i3} \vee y_2) \wedge \cdots \wedge (\bar{y}_{j-2} \vee l_{ij} \vee y_{j-1}) \wedge \cdots \wedge (\bar{y}_{k-4} \vee l_{ik-2} \vee y_{k-3})$
$\wedge (\bar{y}_{k-3} \vee l_{ik-1} \vee l_k)$

We need to prove:

When C_i is satisfiable, the new Z_i is satisfiable

When C_i is not satisfiable, the new Z_i is not satisfiable

When C_i is satisfiable, at least one literal in $\{l_{i1}, l_{i2}, \cdots l_{ik}\}$ is true.

1) If any of l_{i1} or l_{i2} is true, set all the additional variables $y_1, y_2, \cdots, y_{k-3}$ to false.

The first term of all the clauses in Z_i other than the first has a literal $\bar{y}_n$ which evaluates to true.

So Z_i has a satisfying assignment.

$(l_{i1} \vee l_{i2} \vee y_1) \wedge (\bar{y}_1 \vee l_{i3} \vee y_2) \wedge \cdots \wedge (\bar{y}_{j-2} \vee l_{ij} \vee y_{j-1}) \wedge \cdots \wedge (\bar{y}_{k-4} \vee l_{ik-2} \vee y_{k-3}) \wedge (\bar{y}_{k-3} \vee l_{ik-1} \vee l_k)$

2) If any of l_{ik-1} or l_{ik} is true, set all the additional variables $y_1, y_2, \cdots, y_{k-3}$ to true.

The third term of all the clauses in Z_i other than the last has a literal y_n which evaluates to true.

So Z_i has a satisfying assignment.

$(l_{i1} \vee l_{i2} \vee y_1) \wedge (\bar{y}_1 \vee l_{i3} \vee y_2) \wedge \cdots \wedge (\bar{y}_{j-2} \vee l_{ij} \vee y_{j-1}) \wedge \cdots \wedge (\bar{y}_{k-4} \vee l_{ik-2} \vee y_{k-3}) \wedge (\bar{y}_{k-3} \vee l_{ik-1} \vee l_k)$

When C_i is satisfiable, at least one literal in $\{l_{i1}, l_{i2}, \cdots l_{ik}\}$ is true.

3) If l_{ij} (where j $\notin \{1,2,k-1,k\}$) is true, set $y_1, y_2, \cdots, y_{j-2}$ to true and $y_{j-1}, \cdots, y_{k-3}$ to false.

Let us call the clause in Z_i containing l_{ij} as C'

The third term of all the clauses in Z_i left C' has a literal y_n (where $n \in \{1,2,\cdots,j-2\}$)) which evaluates to true.

The first term of all the clauses in Z_i right C' has a literal $\bar{y}_n$ (where $n \in \{j-1,\cdots,k-3,\}$)) which evaluates to true.

So Z_i has a satisfying assignment.

$(l_{i1} \vee l_{i2} \vee y_1) \wedge (\bar{y}_1 \vee l_{i3} \vee y_2) \wedge \cdots \wedge \underbrace{(\bar{y}_{j-2} \vee l_{ij} \vee y_{j-1})}_{C'} \wedge \cdots \wedge (\bar{y}_{k-4} \vee l_{ik-2} \vee y_{k-3}) \wedge (\bar{y}_{k-3} \vee l_{ik-1} \vee l_k)$

When C_i is satisfiable, the new Z_i is satisfiable

When C_i is not satisfiable, no literal in $\{l_{i1}, l_{i2}, \cdots l_{ik}\}$ is true

For Z_i to be satisfiable, all its clauses must evaluate to true.

For the first clause to be true, y_1=true

For the second clause to be true, y_2=true

For the third clause to be true, y_3=true

Similarly for the forth clause to be true, y_4=true

y_4=true $\Rightarrow$ y_5= true $\Rightarrow \cdots \Rightarrow$ y_{k-4}= true$\Rightarrow$ y_{k-3}= true

$(l_{i1} \vee l_{i2} \vee y_1) \wedge (\bar{y}_1 \vee l_{i3} \vee y_2) \wedge \cdots \wedge (\bar{y}_{j-2} \vee l_{ij} \vee y_{j-1}) \wedge \cdots \wedge (\bar{y}_{k-4} \vee l_{ik-2} \vee y_{k-3}) \wedge (\bar{y}_{k-3} \vee l_{ik-1} \vee l_k)$

The last clause evaluates to false . Hence Z_i is not satisfiable.

So we have proved:

When C_i is satisfiable, the new Z_i is satisfiable

When C_i is not satisfiable, the new Z_i is not satisfiable

Hence any clause in a SAT expression can be replaced by a conjunction of clauses which contain 3 literals each.

Therefore, an instance of SAT can be reduced to an equivalent instance of 3-SAT in polynomial time.

c. Deduce that 3-SAT is NP-Complete.

3SAT belonging to the class NP and being NP hard means that 3SAT is NP Complete.